CNN Battle Royale: Presentation Script

Architecture Assigned: VGGNet ("The Deep Worker") — Visual Geometry Group, Oxford

Activity Type: Group Video Presentation **Group Size:** 6 Members **Target Duration:** ~5-6 Minutes

Group Presentation Overview & Structure Mapping

Role	Required Rubric Question	Slide Visual Alignment	Key Topic / Focus
Member 1	1. Who Am I?	Slide 2 (Question 1)	Name, Year (2014), Oxford Creators, AlexNet background
Member 2	2. What Problems Did I Solve?	Slide 3 (Question 2)	Standardized design, 3×3 vs 5×5 math, parameter reduction
Member 3	3. My Superpowers	Slide 4 & Slide 5 (Question 3)	VGG16 Architecture Diagram, 3×3 stacks, Master Craftsman analogy
Member 4	4. My Weaknesses	Slide 6 (Question 4)	138M parameter size, FC layer bottleneck, edge device limits
Member 5	5. Real-World Applications	Slide 7 (Question 5)	Neural Style Transfer, SSD & Faster R-CNN backbones, Medical Imaging
Member 6	6. Why Should We Choose You?	Slide 8 & Slide 9 (Question 6)	Competitor matrix (AlexNet, GoogLeNet, ResNet), final winning pitch

Complete 6-Member Script

Member 1 — Question 1: Who Am I?

Active Visual: Slide 2

> 📷 **Visual Prompt: Display Slide 2 (Question 1: Who Am I?) showing the 2012-2014 timeline, Karen Simonyan & Andrew Zisserman credits, and ILSVRC trophies.**

Member 1: "Welcome to the CNN Battle Royale! Today, our group is defending **VGGNet**, affectionately known as **'The Deep Worker'**."

"To answer our first question, **Who Am I?**:

- **Name:** VGGNet—specifically VGG16 and VGG19, short for Visual Geometry Group Network.
- **Year Developed:** 2014.
- **Creators:** Karen Simonyan and Andrew Zisserman from the Visual Geometry Group at the University of Oxford.
- **Historical Background:** In 2012, AlexNet proved neural networks worked, but used arbitrary, mixed filter sizes—11x11 here, 5x5 there. As shown on the slide's timeline, VGGNet introduced systemized discipline in 2014, proving that *simplicity and uniform depth beat messy design heuristics*. It dominated ILSVRC 2014, winning 1st place in Localization and 2nd in Classification."

Member 2 — Question 2: What Problems Did I Solve?

Active Visual: Slide 3

> 📷 **Visual Prompt: Display Slide 3 (Question 2: What Problems Did I Solve?) showing parameter reduction math ($18C^2$ vs $25C^2$) and dual ReLU activation diagram.**

Member 2: "Moving to question two, **What Problems Did I Solve?**:

Before VGGNet, deep learning was plagued by two major challenges:

1. Inconsistent Architecture Design: Early models guessed where to place different kernel sizes. VGG eliminated this trial-and-error approach by introducing a standard 3x3 block design throughout the entire network.

2. Computational Inefficiency of Large Filters: Researchers thought huge 7x7 or 11x11 filters were needed to see wide visual contexts. As you can see in the parameter formula on screen, VGG proved that stacking **two small 3x3 filters** gives you the exact same spatial coverage as a 5x5 filter, but with **28% fewer parameters** and **double the non-linearity** via two ReLU activations instead of one!"

Member 3 — Question 3: My Superpowers & Fun Analogy

Active Visual: Slide 4 & Slide 5

> **📷 Visual Prompt: Display Slide 4 & Slide 5 (Question 3: My Superpowers & Architecture Diagram) showing the 16-layer block progression and Master Craftsman card.**

Member 3: "Question three: **My Superpowers**.

VGGNet's key strength is its architectural elegance—it uses strictly 3x3 convolutions with stride 1, paired with 2x2 max-pooling across 5 clean feature blocks. On ImageNet 2014, it achieved a staggering **7.3% top-5 error rate**.

(Transition to Slide 5)

Looking at our architecture diagram on screen, you can see how VGG16 systematically expands image features from 64 up to 512 channels across its 16 weight layers.

This brings us to our **Fun Analogy**:

> *"VGGNet is like a Heavy-Duty Master Craftsman. While modern architectures rely on shortcut tricks, VGGNet is an expert artisan who chisels away at an image layer-by-layer using a single, reliable 3x3 tool. It's heavy and takes time, but the resulting craftsmanship is unmatched in quality!"*

Member 4 — Question 4: My Weaknesses

Active Visual: Slide 6

> **📷 Visual Prompt: Display Slide 6 (Question 4: My Weaknesses) showing 138M parameter weight breakdown, FC layer bottleneck callout, and edge device comparison.**

Member 4: "Now for question four: **My Weaknesses**.

Even master craftsmen have limitations. As highlighted on Slide 6, VGGNet's primary drawbacks are:

1. Enormous Model Size: VGG16 has **138 million parameters** and takes over 500 MB of storage space.

2. Memory Bottlenecks: Over **100 million parameters** are crammed into the very first Fully Connected layer alone!

3. Situations Where Other Models Win: On edge devices, smartphones, or robotics where memory and battery life are limited, lightweight models like **MobileNet** or **EfficientNet** perform much better."

Member 5 — Question 5: Real-World Applications

Active Visual: Slide 7

> 📷 **Visual Prompt: Display Slide 7 (Question 5: Real-World Applications) featuring Neural Style Transfer visuals, SSD/Faster R-CNN object detection, and Medical Imaging.**

Member 5: "Question five: **Real-World Applications**.

VGGNet is heavily utilized across multiple industries where visual detail matters most:

- **Digital Art & Creative Tech:** In **Neural Style Transfer** (Gatys et al.), VGG16 is used to extract artistic textures and transfer Van Gogh's brushstrokes onto regular photos.
- **Autonomous Systems & Security:** VGG serves as the core feature extraction backbone in classical object detection frameworks like **SSD (Single Shot MultiBox Detector)** and **Faster R-CNN**.
- **Medical Imaging:** VGG's deep representations are widely fine-tuned for MRI scan classification and tumor detection.

Member 6 — Question 6: Why Should We Choose You?

Active Visual: Slide 8 & Slide 9

> 📷 **Visual Prompt: Display Slide 8 & Slide 9 (Question 6: Why Should We Choose You?) showing the competitor matrix (AlexNet, GoogLeNet, ResNet) and the final pitch.**

Member 6: "Finally, question six: **Why Should We Choose You?**

As you can see in our side-by-side comparison matrix on Slide 8:

- **AlexNet** is an outdated relic with crude, arbitrary filter choices.
- **GoogLeNet** is over-engineered with complex, hard-to-tune multi-branch Inception blocks.
- While **ResNet** added skip connections later, **VGGNet is the true pioneer that proved depth and uniformity are what make neural networks intelligent**.

(Transition to Slide 9)

VGGNet provided the foundational blueprint that made modern deep computer vision possible. If you want the most predictable, reliable, and high-fidelity feature extractor ever created—you choose **VGGNet: The Deep Worker**! Thank you!"